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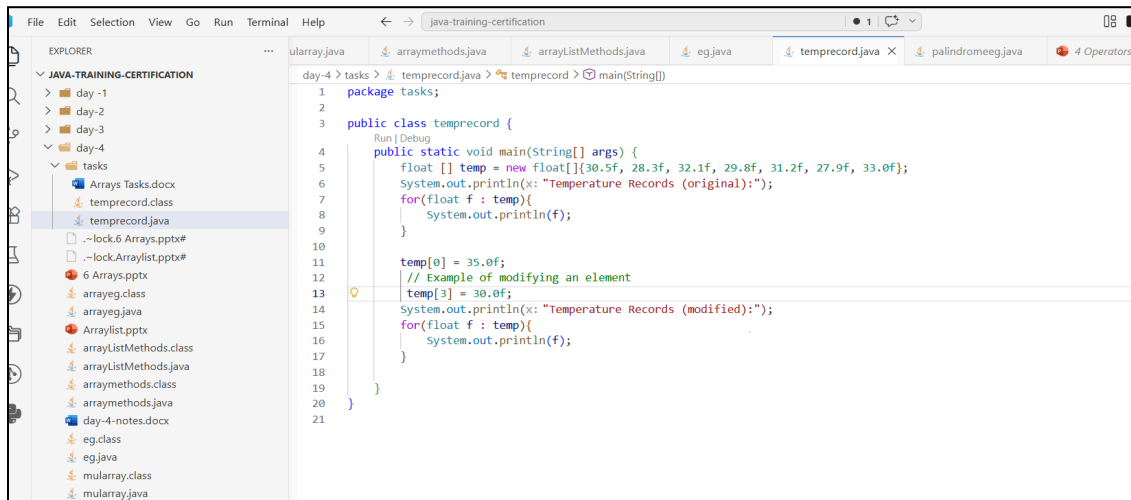
Branch: Bsc Information systems

College: Kongu Engineering College

Task: Array Tasks

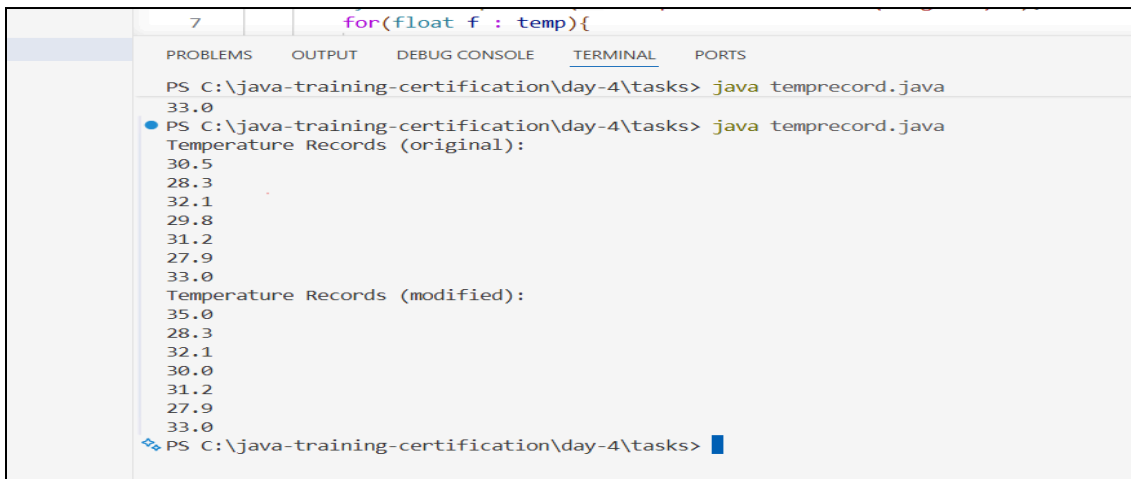
Date: 07/05/2026

## 1. TemperatureRecord.java



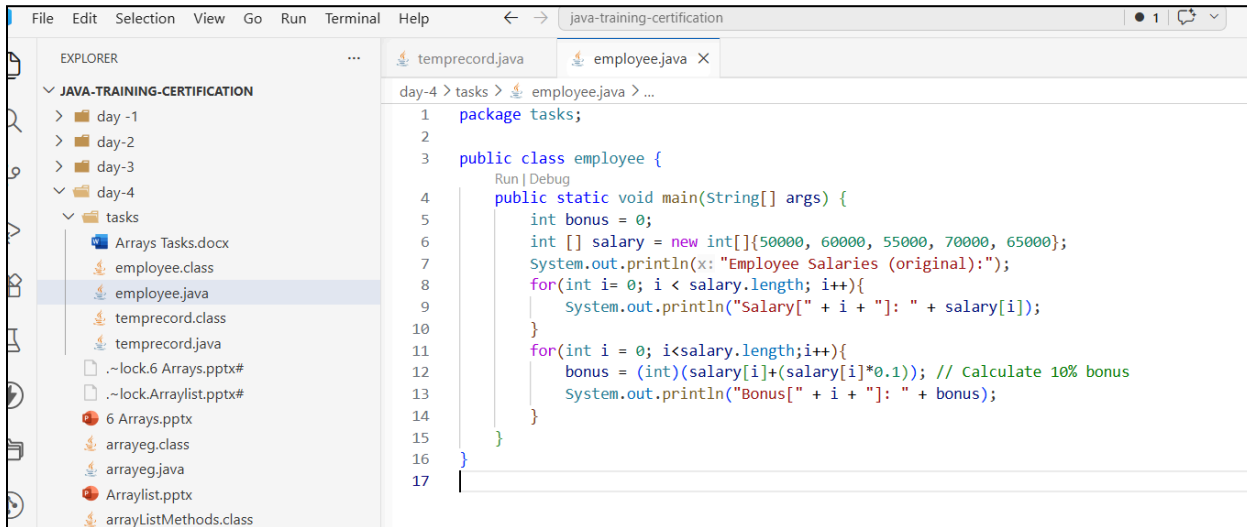
```
1 package tasks;
2
3 public class temprecord {
4     public static void main(String[] args) {
5         float [] temp = new float[]{30.5f, 28.3f, 32.1f, 29.8f, 31.2f, 27.9f, 33.0f};
6         System.out.println(x: "Temperature Records (original):");
7         for(float f : temp){
8             System.out.println(f);
9         }
10
11         temp[0] = 35.0f;
12         // Example of modifying an element
13         temp[3] = 30.0f;
14         System.out.println(x: "Temperature Records (modified):");
15         for(float f : temp){
16             System.out.println(f);
17         }
18     }
19 }
20
21 }
```

## Output:



```
PS C:\java-training-certification\day-4\tasks> java temprecord.java
33.0
PS C:\java-training-certification\day-4\tasks> java temprecord.java
Temperature Records (original):
30.5
28.3
32.1
29.8
31.2
27.9
33.0
Temperature Records (modified):
35.0
28.3
32.1
30.0
31.2
27.9
33.0
PS C:\java-training-certification\day-4\tasks>
```

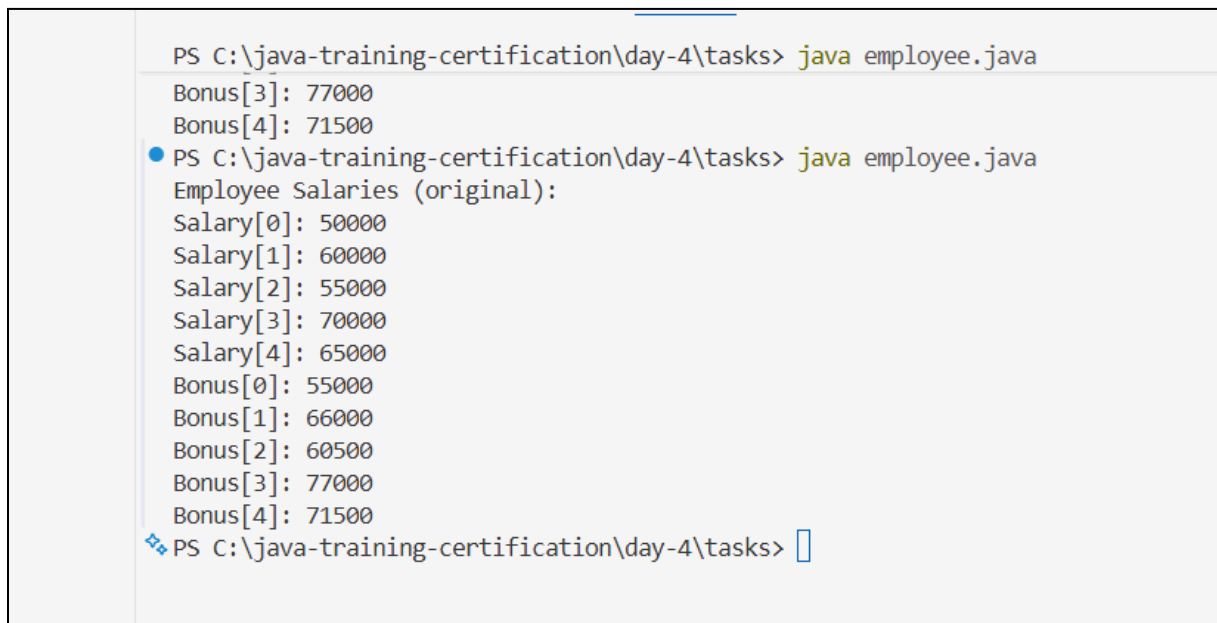
## 2. Employee Bonus Calculator



The screenshot shows an IDE with the Explorer pane on the left displaying the project structure. The main editor shows the code for `employee.java`. The code defines a package `tasks` and a class `employee` with a `main` method. The `main` method initializes an array of salaries, prints them, and then calculates a 10% bonus for each, printing the results.

```
1 package tasks;
2
3 public class employee {
4     public static void main(String[] args) {
5         int bonus = 0;
6         int [] salary = new int[]{50000, 60000, 55000, 70000, 65000};
7         System.out.println("Employee Salaries (original):");
8         for(int i= 0; i < salary.length; i++){
9             System.out.println("Salary[" + i + "]: " + salary[i]);
10        }
11        for(int i = 0; i<salary.length;i++){
12            bonus = (int)(salary[i]+(salary[i]*0.1)); // Calculate 10% bonus
13            System.out.println("Bonus[" + i + "]: " + bonus);
14        }
15    }
16 }
17
```

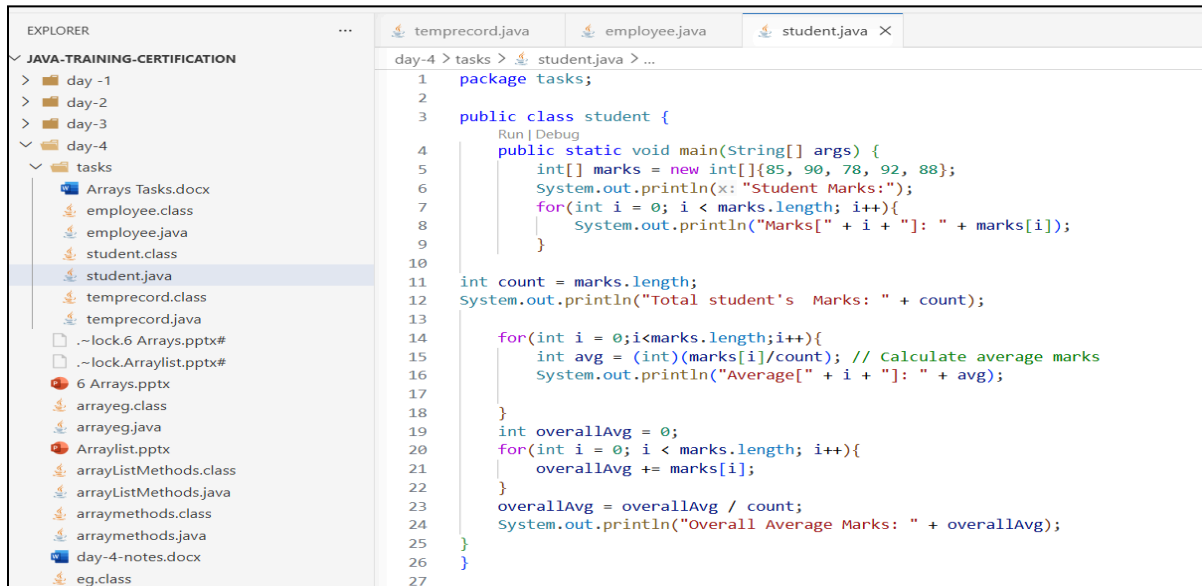
## Output



The screenshot shows a terminal window with the following output:

```
PS C:\java-training-certification\day-4\tasks> java employee.java
Bonus[3]: 77000
Bonus[4]: 71500
PS C:\java-training-certification\day-4\tasks> java employee.java
Employee Salaries (original):
Salary[0]: 50000
Salary[1]: 60000
Salary[2]: 55000
Salary[3]: 70000
Salary[4]: 65000
Bonus[0]: 55000
Bonus[1]: 66000
Bonus[2]: 60500
Bonus[3]: 77000
Bonus[4]: 71500
PS C:\java-training-certification\day-4\tasks>
```

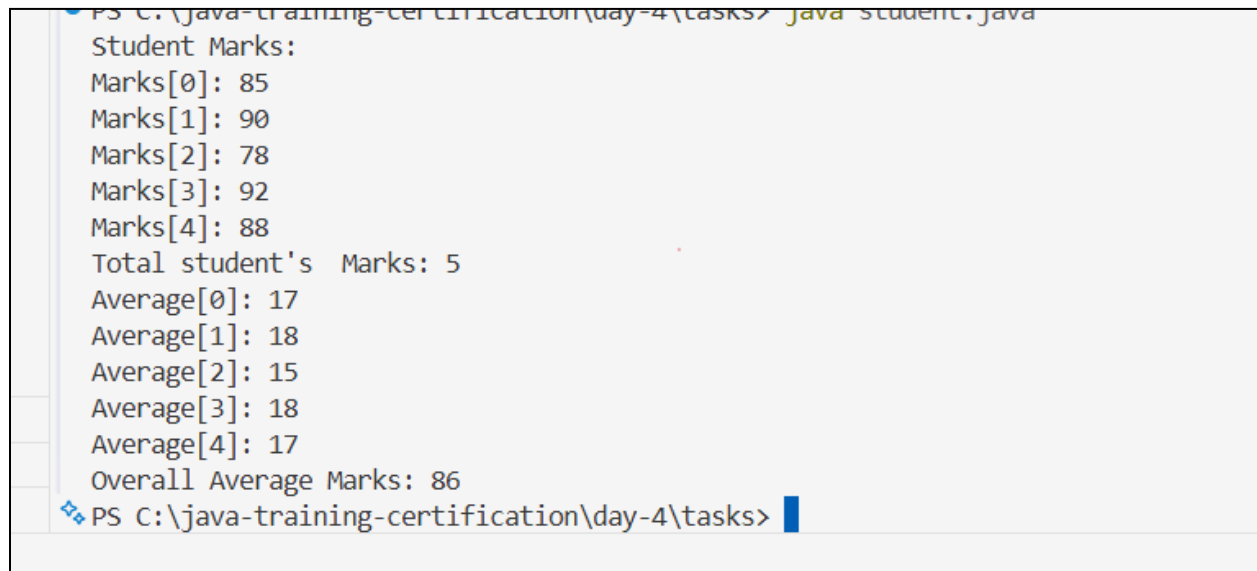
### 3. Student Marks Average



The screenshot shows an IDE with a project named 'JAVA-TRAINING-CERTIFICATION'. The 'EXPLORER' pane on the left shows the file structure, with 'student.java' selected under 'day-4/tasks'. The main editor displays the code for 'student.java'.

```
1 package tasks;
2
3 public class student {
4     public static void main(String[] args) {
5         int[] marks = new int[]{85, 90, 78, 92, 88};
6         System.out.println("Student Marks:");
7         for(int i = 0; i < marks.length; i++){
8             System.out.println("Marks[" + i + "]: " + marks[i]);
9         }
10
11         int count = marks.length;
12         System.out.println("Total student's Marks: " + count);
13
14         for(int i = 0; i < marks.length; i++){
15             int avg = (int)(marks[i]/count); // Calculate average marks
16             System.out.println("Average[" + i + "]: " + avg);
17         }
18
19         int overallAvg = 0;
20         for(int i = 0; i < marks.length; i++){
21             overallAvg += marks[i];
22         }
23         overallAvg = overallAvg / count;
24         System.out.println("Overall Average Marks: " + overallAvg);
25     }
26 }
27
```

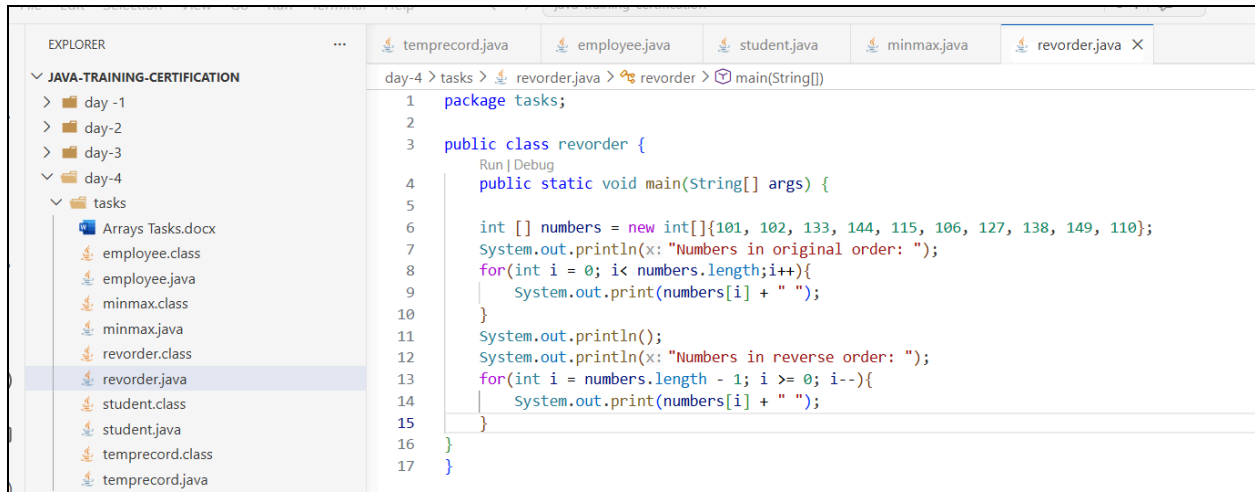
### Output:



The screenshot shows a command prompt window with the following output:

```
PS C:\java-training-certification\day-4\tasks> java student.java
Student Marks:
Marks[0]: 85
Marks[1]: 90
Marks[2]: 78
Marks[3]: 92
Marks[4]: 88
Total student's Marks: 5
Average[0]: 17
Average[1]: 18
Average[2]: 15
Average[3]: 18
Average[4]: 17
Overall Average Marks: 86
PS C:\java-training-certification\day-4\tasks>
```

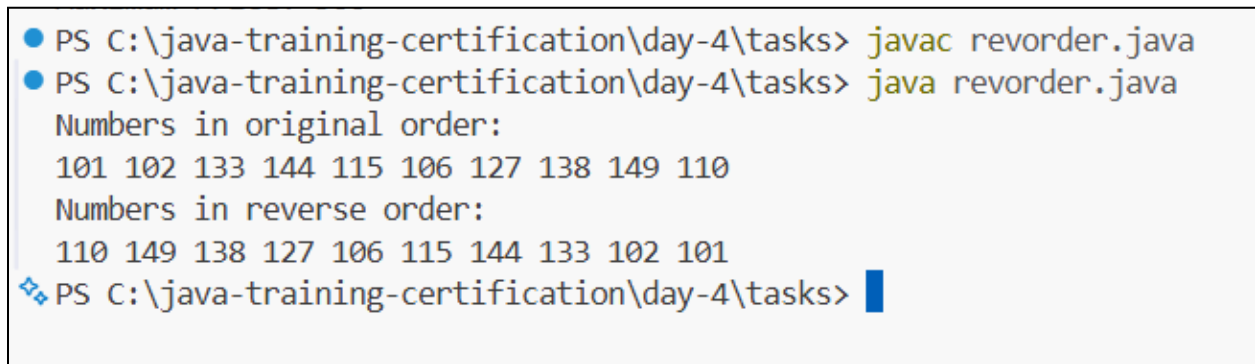
#### 4. Reverse order of an array:



The screenshot shows an IDE with a project named 'JAVA-TRAINING-CERTIFICATION'. The 'EXPLORER' pane on the left shows a folder structure with 'day-4' containing a 'tasks' folder. The 'tasks' folder contains several files, including 'revorder.java'. The 'REVIEWER' pane on the right shows the code for 'revorder.java'. The code defines a package 'tasks' and a public class 'revorder'. The 'main' method prints the original order of an array of integers and then prints the array in reverse order.

```
1 package tasks;
2
3 public class revorder {
4     public static void main(String[] args) {
5
6         int [] numbers = new int[]{101, 102, 133, 144, 115, 106, 127, 138, 149, 110};
7         System.out.println("Numbers in original order: ");
8         for(int i = 0; i < numbers.length; i++){
9             System.out.print(numbers[i] + " ");
10        }
11        System.out.println();
12        System.out.println("Numbers in reverse order: ");
13        for(int i = numbers.length - 1; i >= 0; i--){
14            System.out.print(numbers[i] + " ");
15        }
16    }
17 }
```

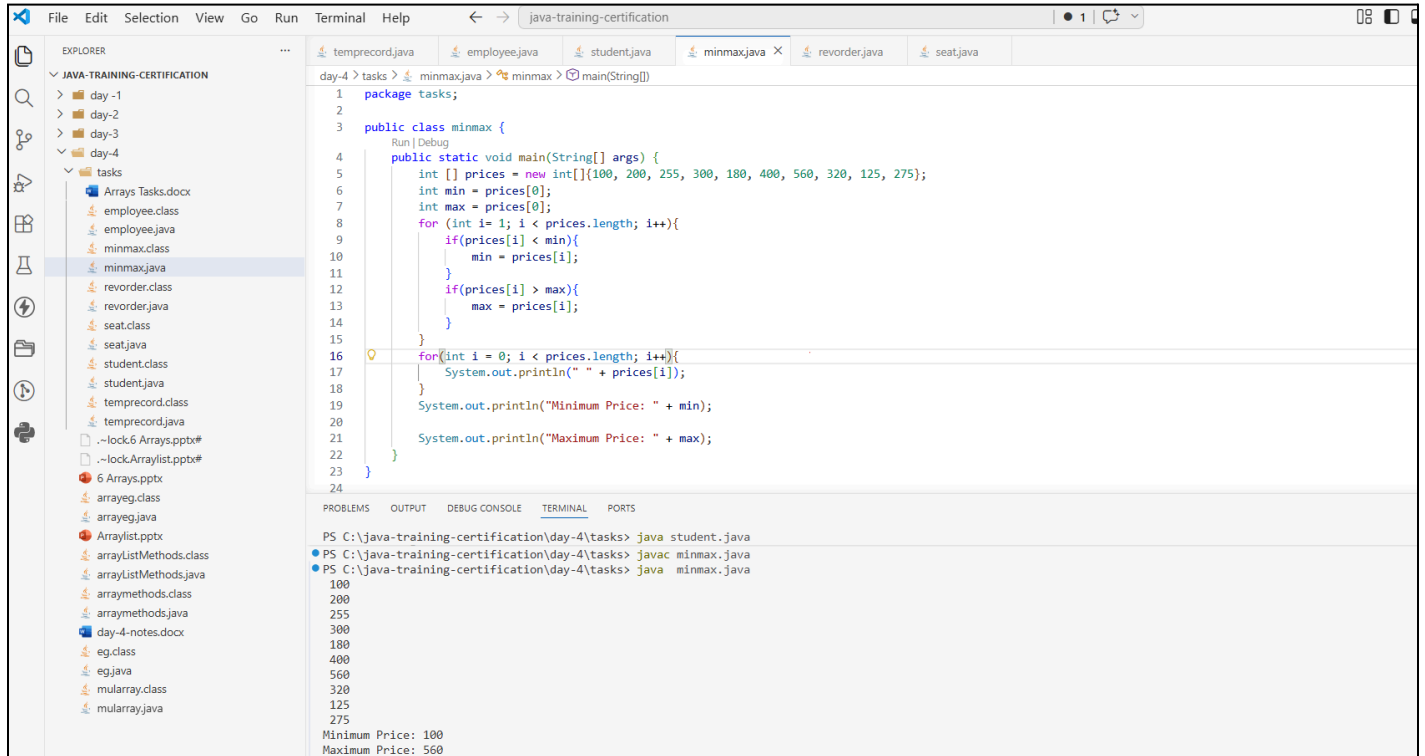
#### Output:



The screenshot shows a command prompt window with the following commands and output:

```
PS C:\java-training-certification\day-4\tasks> javac revorder.java
PS C:\java-training-certification\day-4\tasks> java revorder.java
Numbers in original order:
101 102 133 144 115 106 127 138 149 110
Numbers in reverse order:
110 149 138 127 106 115 144 133 102 101
PS C:\java-training-certification\day-4\tasks>
```

## 5. Find Maximum and Minimum Price



The screenshot shows an IDE window titled "java-training-certification". The Explorer panel on the left shows a project structure with folders "day-1", "day-2", "day-3", and "day-4". Under "day-4", there is a folder "tasks" containing several files, including "minmax.java" which is currently selected. The main editor displays the code for "minmax.java". The code defines a package "tasks", a class "minmax", and a static method "main" that takes a String array "args". Inside "main", an array of integers "prices" is initialized with values: 100, 200, 255, 300, 180, 400, 560, 320, 125, 275. The code then iterates through the array to find the minimum and maximum values. The output of the program is shown in the Terminal panel at the bottom.

```
1 package tasks;
2
3 public class minmax {
4     public static void main(String[] args) {
5         int [] prices = new int[]{100, 200, 255, 300, 180, 400, 560, 320, 125, 275};
6         int min = prices[0];
7         int max = prices[0];
8         for (int i = 1; i < prices.length; i++){
9             if(prices[i] < min){
10                 min = prices[i];
11             }
12             if(prices[i] > max){
13                 max = prices[i];
14             }
15         }
16         for(int i = 0; i < prices.length; i++){
17             System.out.println(" " + prices[i]);
18         }
19         System.out.println("Minimum Price: " + min);
20         System.out.println("Maximum Price: " + max);
21     }
22 }
23
24
```

Terminal Output:

```
PS C:\java-training-certification\day-4\tasks> java student.java
PS C:\java-training-certification\day-4\tasks> javac minmax.java
PS C:\java-training-certification\day-4\tasks> java minmax.java
100
200
255
300
180
400
560
320
125
275
Minimum Price: 100
Maximum Price: 560
```

## 6. Classroom Seat Number Matrix

The screenshot displays an IDE with the following components:

- EXPLORER:** Shows a project structure for 'JAVA-TRAINING-CERTIFICATION' with folders 'day-1', 'day-2', 'day-3', and 'day-4'. The 'tasks' folder under 'day-4' is expanded, listing files like 'Arrays Tasks.docx', 'employee.class', 'employee.java', 'minmax.class', 'minmax.java', 'revorder.class', 'revorder.java', 'seat.class', 'seat.java', 'student.class', 'student.java', 'temprecord.class', and 'temprecord.java'.
- Editor:** Displays the code for 'seat.java' with the following content:

```
1 package tasks;
2
3 public class seat {
4     public static void main(String[] args) {
5         int [] rownum = new int[] {1,2,3,4};
6         int [] seatnum = new int[] {1,2,3};
7         for(int i = 0; i < rownum.length; i++){
8             for(int j = 0; j < seatnum.length; j++){
9                 System.out.println("Row " + rownum[i] + " Seat " + seatnum[j]);
10            }
11        }
12    }
13 }
14
```
- TERMINAL:** Shows the execution of the program and a compilation error.
  - Command: `PS C:\java-training-certification\day-4\tasks> java revorder.java`
  - Command: `PS C:\java-training-certification\day-4\tasks> javac seat.jav`
  - Error: `error: Class names, 'seat.jav', are only accepted if annotation processing is explicitly requested`
  - Command: `PS C:\java-training-certification\day-4\tasks> javac seat.java`
  - Command: `PS C:\java-training-certification\day-4\tasks> java seat.java`
  - Output:

```
Row 1 Seat 1
Row 1 Seat 2
Row 1 Seat 3
Row 2 Seat 1
Row 2 Seat 2
Row 2 Seat 3
Row 3 Seat 1
Row 3 Seat 2
Row 3 Seat 3
Row 4 Seat 1
Row 4 Seat 2
Row 4 Seat 3
```

## 7. Matrix addition:

The screenshot displays an IDE window titled "java-training-certification". The Explorer panel on the left shows a project structure with folders "day-1" through "day-4". Under "day-4", there is a "tasks" folder containing several files, including "add.java" which is currently selected. The main editor area shows the code for "add.java":

```
1 package tasks;
2
3 public class add {
4     public static void main(String[] args) {
5         int [] arr1 = new int[]{1, 2, 3};
6         int [] arr2 = new int[]{4, 5, 6};
7         int [] sum = new int[arr1.length];
8         for(int i = 0; i < arr1.length; i++){
9             sum[i] = arr1[i] + arr2[i];
10        }
11        System.out.println(x: "Sum of arrays: ");
12        for(int i = 0; i < sum.length; i++){
13            System.out.print(sum[i] + " ");
14        }
15        System.out.println();
16    }
17 }
18
```

Below the code editor, the TERMINAL panel shows the execution of the program:

```
PS C:\java-training-certification\day-4\tasks> java seat.java
Row 2 Seat 3
Row 3 Seat 1
Row 3 Seat 2
Row 3 Seat 3
Row 4 Seat 1
Row 4 Seat 2
Row 4 Seat 3
● PS C:\java-training-certification\day-4\tasks> javac add.java
● PS C:\java-training-certification\day-4\tasks> java add.java
Sum of arrays:
5 7 9
❖ PS C:\java-training-certification\day-4\tasks>
```

## 8. Transpose of a matrix:

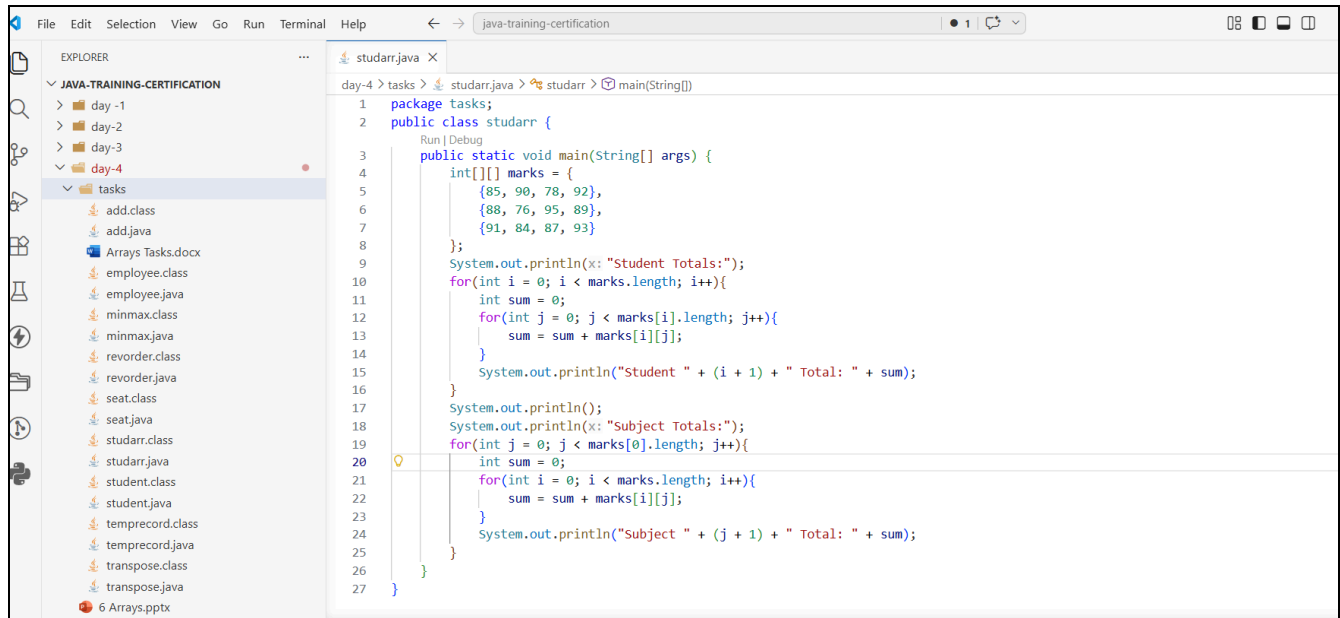
```
1 package tasks;
2
3 public class transpose {
4     Run | Debug
5     public static void main(String[] args) {
6         int[][] matrix = {
7             {1, 2, 3, 0},
8             {4, 5, 6, 0},
9             {7, 8, 9, 0}
10        };
11        int rows = matrix.length;
12        int cols = matrix[0].length;
13        int[][] transpose = new int[cols][rows];
14        for(int i = 0; i < rows; i++){
15            for(int j = 0; j < cols; j++){
16                transpose[j][i] = matrix[i][j];
17            }
18        }
19        System.out.println(x: "Original matrix:");
20        for(int i = 0; i < rows; i++){
21            for(int j = 0; j < cols; j++){
22                System.out.print(matrix[i][j] + " ");
23            }
24            System.out.println();
25        }
26        System.out.println(x: "Transposed matrix:");
27        for(int i = 0; i < cols; i++){
28            for(int j = 0; j < rows; j++){
29                System.out.print(transpose[i][j] + " ");
30            }
31            System.out.println();
32        }
33    }
34 }
```



Output:

```
PS C:\java-training-certification\day-4\tasks> javac transpose.java
PS C:\java-training-certification\day-4\tasks> java transpose.java
Original matrix:
1 2 3 0
4 5 6 0
7 8 9 0
Transposed matrix:
1 4 7
2 5 8
3 6 9
0 0 0
PS C:\java-training-certification\day-4\tasks> 
```

## 9. Student Marks per subject



The screenshot shows an IDE with the Explorer panel on the left displaying a project structure for 'JAVA-TRAINING-CERTIFICATION'. The 'tasks' folder under 'day-4' is selected, showing files like 'add.class', 'add.java', 'ArraysTasks.docx', 'employee.class', 'employee.java', 'minmax.class', 'minmax.java', 'revorder.class', 'revorder.java', 'seat.class', 'seat.java', 'studarr.class', 'studarr.java', 'student.class', 'student.java', 'temprecord.class', 'temprecord.java', 'transpose.class', and 'transpose.java'. The main editor displays the code for 'studarr.java'.

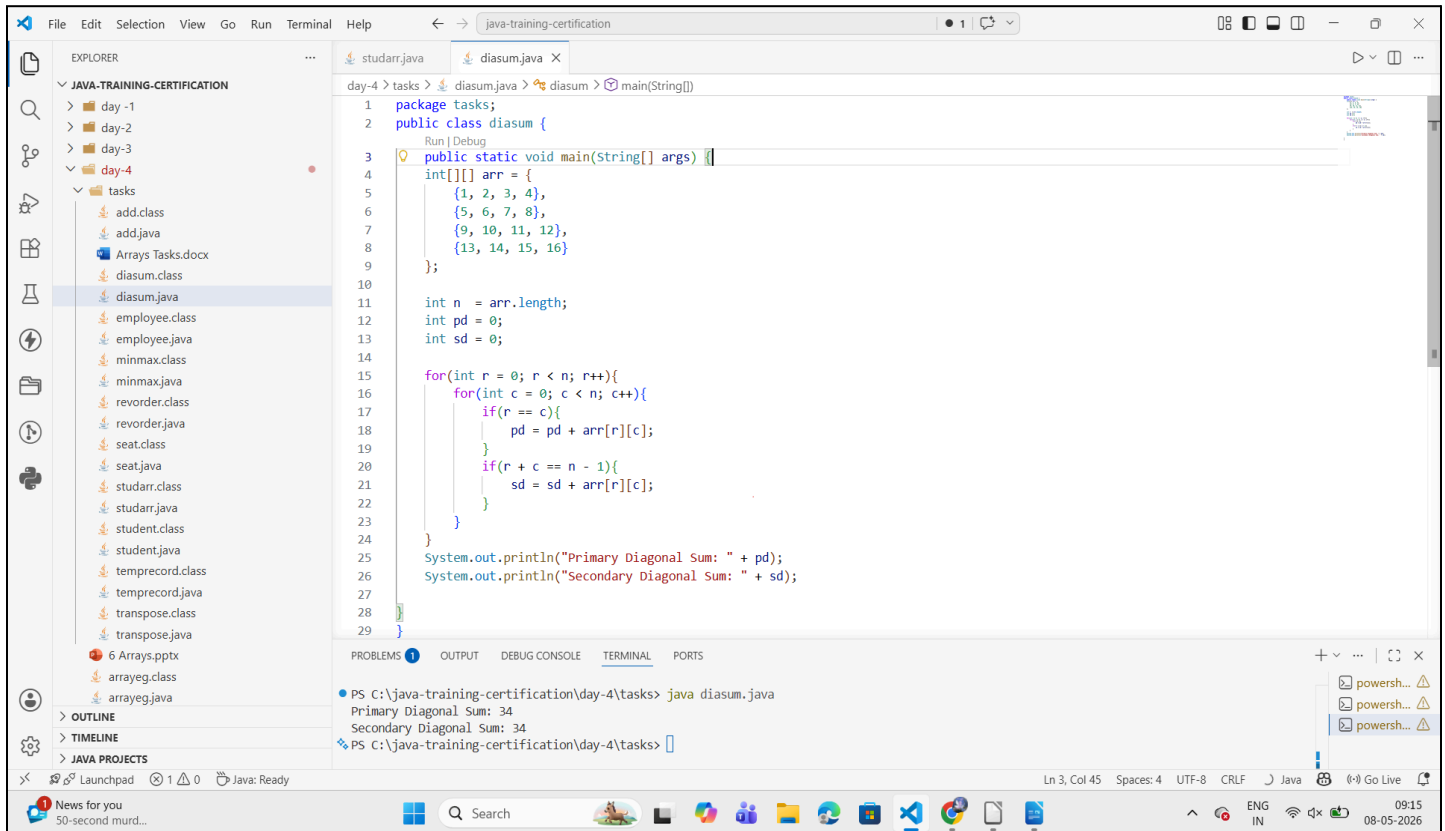
```
1 package tasks;
2 public class studarr {
3     public static void main(String[] args) {
4         int[][] marks = {
5             {85, 90, 78, 92},
6             {88, 76, 95, 89},
7             {91, 84, 87, 93}
8         };
9         System.out.println("Student Totals:");
10        for(int i = 0; i < marks.length; i++){
11            int sum = 0;
12            for(int j = 0; j < marks[i].length; j++){
13                sum = sum + marks[i][j];
14            }
15            System.out.println("Student " + (i + 1) + " Total: " + sum);
16        }
17        System.out.println();
18        System.out.println("Subject Totals:");
19        for(int j = 0; j < marks[0].length; j++){
20            int sum = 0;
21            for(int i = 0; i < marks.length; i++){
22                sum = sum + marks[i][j];
23            }
24            System.out.println("Subject " + (j + 1) + " Total: " + sum);
25        }
26    }
27 }
```

### Output:

```
PS C:\java-training-certification> cd day-4
PS C:\java-training-certification\day-4> cd tasks
PS C:\java-training-certification\day-4\tasks> javac studarr.java
PS C:\java-training-certification\day-4\tasks> java studarr.java
Student Totals:
Student 1 Total: 345
Student 2 Total: 348
Student 3 Total: 355

Subject Totals:
Subject 1 Total: 264
Subject 2 Total: 250
Subject 3 Total: 260
Subject 4 Total: 274
PS C:\java-training-certification\day-4\tasks>
```

## 10. Diagonal sum of a matrix



The screenshot shows an IDE window titled "java-training-certification". The Explorer panel on the left shows a project structure with folders "day-1", "day-2", "day-3", and "day-4". Under "day-4", there is a "tasks" folder containing several files, including "diasum.java" which is currently selected. The main editor displays the code for "diasum.java". The code defines a package "tasks", a public class "diasum", and a main method. It initializes a 4x4 integer array "arr" with the following values:

```
1 package tasks;
2 public class diasum {
3     public static void main(String[] args) {
4         int[][] arr = {
5             {1, 2, 3, 4},
6             {5, 6, 7, 8},
7             {9, 10, 11, 12},
8             {13, 14, 15, 16}
9         };
10
11         int n = arr.length;
12         int pd = 0;
13         int sd = 0;
14
15         for(int r = 0; r < n; r++){
16             for(int c = 0; c < n; c++){
17                 if(r == c){
18                     pd = pd + arr[r][c];
19                 }
20                 if(r + c == n - 1){
21                     sd = sd + arr[r][c];
22                 }
23             }
24         }
25         System.out.println("Primary Diagonal Sum: " + pd);
26         System.out.println("Secondary Diagonal Sum: " + sd);
27     }
28 }
29
```

The output window at the bottom shows the results of running the program:

```
PS C:\java-training-certification\day-4\tasks> java diasum.java
Primary Diagonal Sum: 34
Secondary Diagonal Sum: 34
PS C:\java-training-certification\day-4\tasks>
```

The status bar at the bottom indicates the cursor is at line 3, column 45, with 4 spaces, UTF-8 encoding, and CRLF line endings. The system tray shows the date and time as 09:15 on 08-05-2026.























